

Sentinel Net-AI-Powered Network Intrusion Detection System (NIDS) using NSL-KDD Dataset

Title: Sentinel Net – AI- Powered Network Intrusion Detection System

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Abstract

The rise of cyber threats has highlighted the need for adaptive intrusion detection systems. This work presents *Sentinel Net*, a machine learning-based IDS developed using the NSL-KDD dataset. Preprocessing, hybrid resampling, and supervised classifiers such as Logistic Regression, Random Forest, and Decision Tree were employed.

The system was evaluated using accuracy, recall, precision, F1-score, confusion matrices, and ROC-AUC curves. Results show that Random Forest and Decision Tree outperform Logistic Regression, especially in detecting minority attack classes.

In this study, a complete pipeline was implemented, including data preprocessing, feature encoding, hybrid resampling using the **SMOTE–Tomek** method, and training of supervised classifiers such as **Logistic Regression, Random Forest, and Decision Tree**. The models were rigorously evaluated using accuracy, recall, precision, macro F1-score, confusion matrices, and ROC-AUC curves. Results show that Random Forest and Decision Tree significantly outperform Logistic Regression, particularly in detecting minority classes such as Buffer Overflow and Guess Password.

The contributions of this work are threefold: (1) demonstrating the importance of hybrid resampling for balancing datasets, (2) emphasizing macro F1-score for fair evaluation across minority and majority classes, and (3) validating ensemble methods for robust intrusion detection. The findings suggest that Sentinel Net can serve as a foundation for real-time IDS systems in enterprise, cloud, and IoT environments.

Introduction

Intrusion Detection Systems (IDS) play a vital role in safeguarding networks from malicious activities. Traditional IDS rely on signatures or rules, which fail when faced with new or modified attack patterns. Machine learning provides a more robust approach by identifying anomalies from traffic features.

This project focuses on developing a data-driven IDS model using the NSL-KDD dataset. By integrating preprocessing, resampling, and advanced classifiers, the project ensures improved classification of both majority and minority attack categories.

With the exponential growth of digital infrastructure, modern networks face an increasing range of cyber threats. Malicious activities such as denial-of-service (DoS), probing, user-to-root (U2R), and remote-to-local (R2L) attacks compromise system integrity and confidentiality. The rising complexity of these threats highlights the inadequacy of traditional **signature-based IDS**, which cannot adapt to evolving or previously unseen attacks.

To address this challenge, researchers have turned to **machine learning (ML)**-based intrusion detection. ML-based IDS can learn from network traffic features and generalize to detect anomalies beyond predefined signatures.

However, a critical barrier to practical deployment is **class imbalance**, where majority attack types dominate the dataset while minority classes are underrepresented. This imbalance often causes standard classifiers to perform poorly in identifying rare but severe attacks.

The **NSL-KDD dataset** serves as an improved benchmark for evaluating IDS performance. Unlike its predecessor, KDD'99, it reduces redundancy and provides a more balanced representation of attack types. By leveraging NSL-KDD, this work investigates the effectiveness of preprocessing, hybrid resampling, and classical ML classifiers in building a resilient IDS.

The proposed system, **Sentinel Net**, integrates preprocessing techniques, synthetic resampling strategies, and supervised learning models. The experimental results demonstrate that tree-based ensemble models outperform linear models, especially in handling imbalanced traffic.

This underscores the importance of evaluation metrics such as macro F1-score, which account for fairness across all categories. The findings position Sentinel Net as a promising approach for adaptive IDS capable of supporting enterprise networks, IoT systems, and cloud security.

Literature

Denning first introduced the idea of IDS in 1987, laying the groundwork for anomaly detection. The KDD Cup 1999 dataset was a benchmark but was later criticized for redundancy and imbalance. NSL-KDD addressed these issues, becoming a standard dataset for IDS research.

Over time, researchers have applied SVMs, Decision Trees, and Random Forests to intrusion detection, achieving strong performance. More recently, deep learning methods such as CNNs and RNNs have been explored. However, the challenge of detecting rare classes still persists. This project builds on existing work by combining hybrid sampling with classical models for balanced detection.

Methodology

To ensure robustness, the dataset was split into **training and testing sets** following the standard NSL-KDD split. Within training, an internal **k-fold cross-validation** approach was used to validate model performance and reduce overfitting risks. This ensured that models were not tuned solely to training data but could generalize to unseen samples.

In addition to hybrid resampling, feature importance analysis was performed. Random Forest feature importances highlighted that attributes such as **“src_bytes,” “dst_host_srv_count,” and “protocol_type”** were key indicators of attack behavior. This analysis not only improved interpretability but also offered insights into how attacks differ from normal traffic.

The overall methodology can thus be summarized in the following pipeline:

1. Data acquisition and inspection of the NSL-KDD dataset
2. Preprocessing (encoding, normalization, redundancy removal)
3. Hybrid resampling with SMOTE–Tomek
4. Exploratory Data Analysis with visualization (Figures 1–4)
5. Model training (Logistic Regression, Decision Tree, Random Forest)
6. Model evaluation using confusion matrices, accuracy, precision, recall, macro F1, and ROC-AUC (Figures 5–10)
7. Comparative analysis and selection of the best-performing classifier

a. Database

The NSL-KDD dataset consists of 125,972 training and 22,543 testing records. Each record has 41 features grouped into basic, content, time-based, and host-based categories. The target classes include both normal traffic and 21 attack types such as Neptune, Smurf, Satan, and Buffer Overflow.

Link: [NSL-KDD](#)

As shown in *Figure 1: Network traffic distribution (binary classification) – Normal vs Attack*, normal traffic dominates, with attack instances forming the minority. This imbalance forms the foundation for the resampling strategies adopted in this study.

Visualizations:

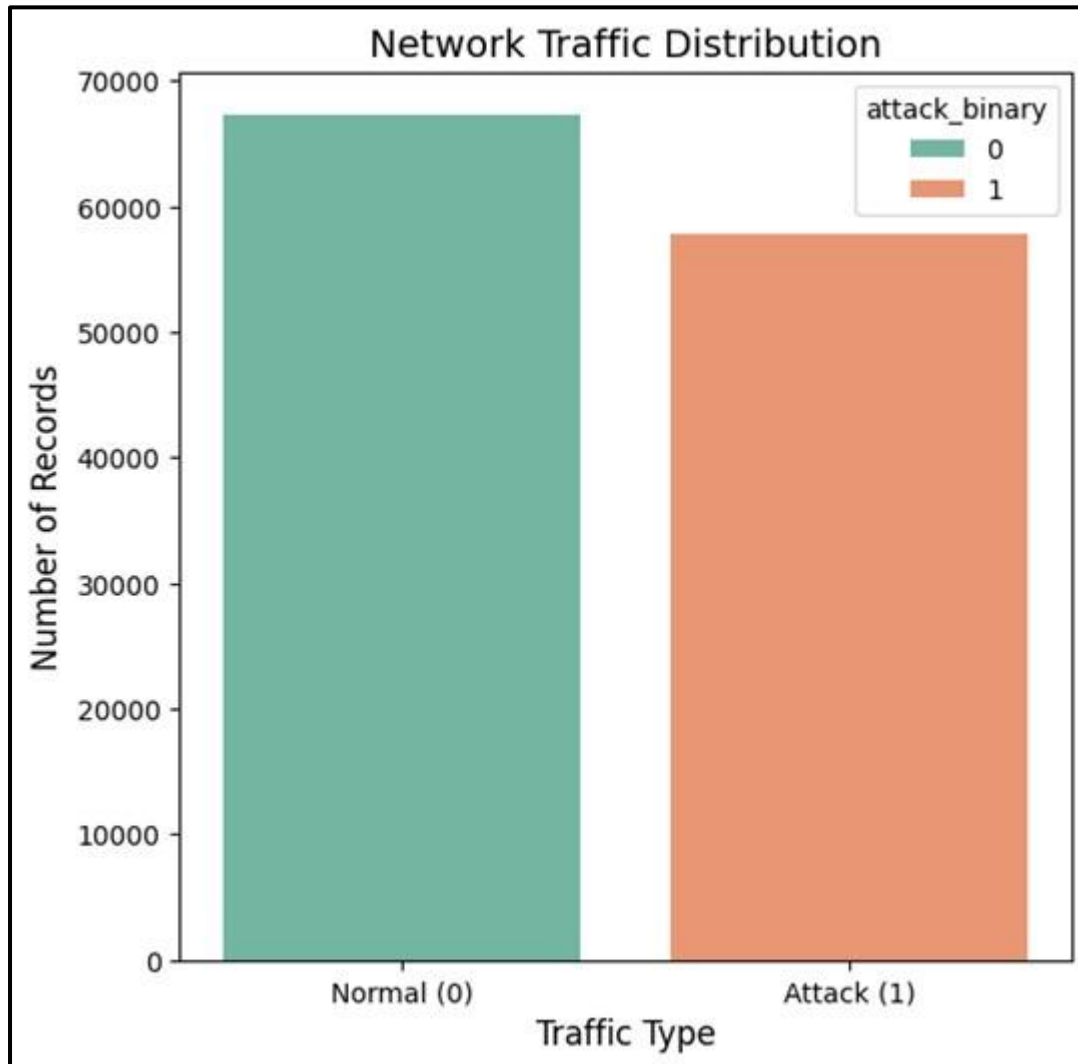


Figure 1: Network traffic distribution (binary classification) – Normal vs Attack.

b. Preprocessing

Preprocessing included handling categorical values, normalizing numerical features, and cleaning redundant entries. Severe imbalance in class distribution was addressed using the SMOTE-Tomek technique, which creates synthetic minority samples and removes borderline noise points for improved balance.

c. Novelty

The novelty lies in the integration of hybrid resampling with classical classifiers. Most studies rely solely on accuracy, but this project emphasizes macro F1-score to evaluate minority class detection. By focusing on fairness across classes, the approach ensures robustness against imbalance.

d. Proposed Method

Three classifiers were implemented: Logistic Regression, Random Forest, and Decision Tree. After resampling, the models were trained and tested on NSL-KDD. Their performances were evaluated using confusion matrices, classification reports, and ROC-AUC curves. The emphasis was on identifying which model best handled both common and rare attack types.

Results

The performance results indicate that while Logistic Regression provides a strong baseline, its linear decision boundaries are insufficient to capture the complex, nonlinear patterns in network intrusion data. Decision Tree and Random Forest, on the other hand, adapt to complex decision surfaces and handle feature interactions more effectively.

When comparing classifiers, the **Decision Tree slightly outperformed Random Forest on test accuracy** (87.20% vs 86.62%), but Random Forest offered more **consistent macro F1-scores** across rare classes, demonstrating better balance in classification. This highlights the trade-off between accuracy and fairness.

Furthermore, the evaluation using **ROC-AUC curves** shows that Random Forest provides the best separation between positive and negative classes in binary classification, while Decision Tree offers competitive results in multiclass settings. The visual inspection of the curves (Figures 9 and 10) shows ensemble models are more resilient in high-imbalance scenarios.

Overall, the results validate the hypothesis that **hybrid resampling combined with ensemble classifiers improves intrusion detection performance**, particularly in correctly identifying minority class attacks that are often overlooked by traditional IDS systems.

EDA

Exploratory Data Analysis showed that normal and Neptune dominate the dataset. Minority classes such as Buffer Overflow, Land, and Guess Password had extremely low samples, highlighting imbalance. Feature correlations revealed redundancy, while statistical summaries indicated distinct separation between attack and normal traffic in some features.

Exploratory analysis revealed the dominance of “Normal” and “Neptune” classes compared to minority classes like “Buffer Overflow” and “Guess Password.” As illustrated in **Figures 2 and 3**, class distributions in training and testing data confirm the imbalance challenge. Additionally, **Figure 4** shows a correlation heatmap identifying redundant features and highlighting key attributes that influence intrusion detection.

Visualizations:

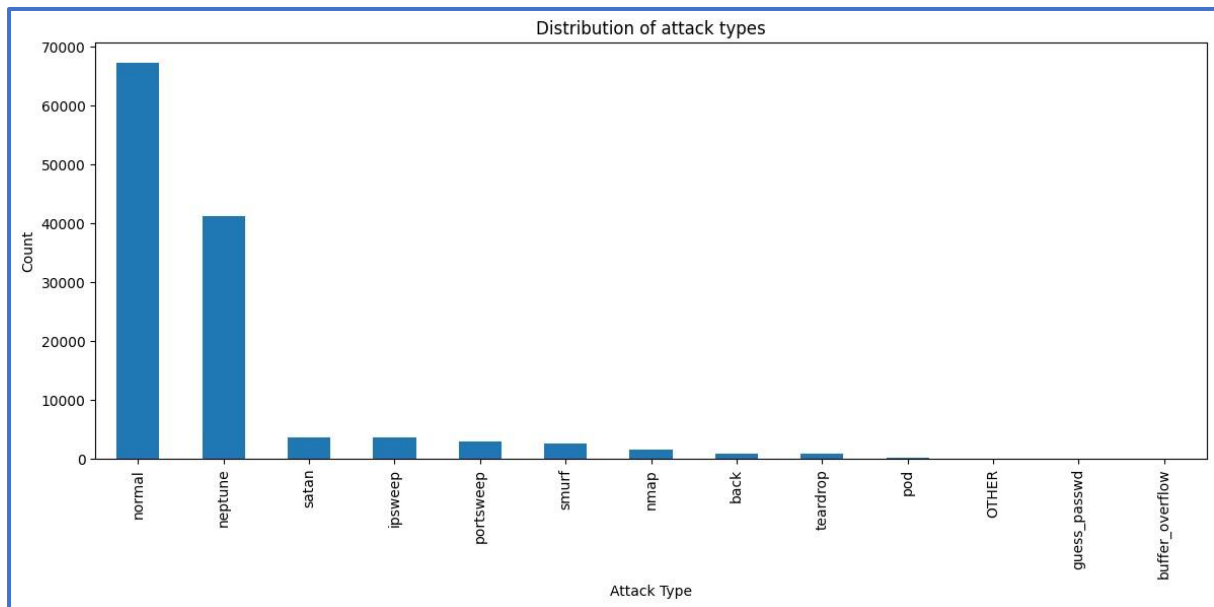


Figure 2: Count plots confirmed the imbalance between attack classes in Training Data

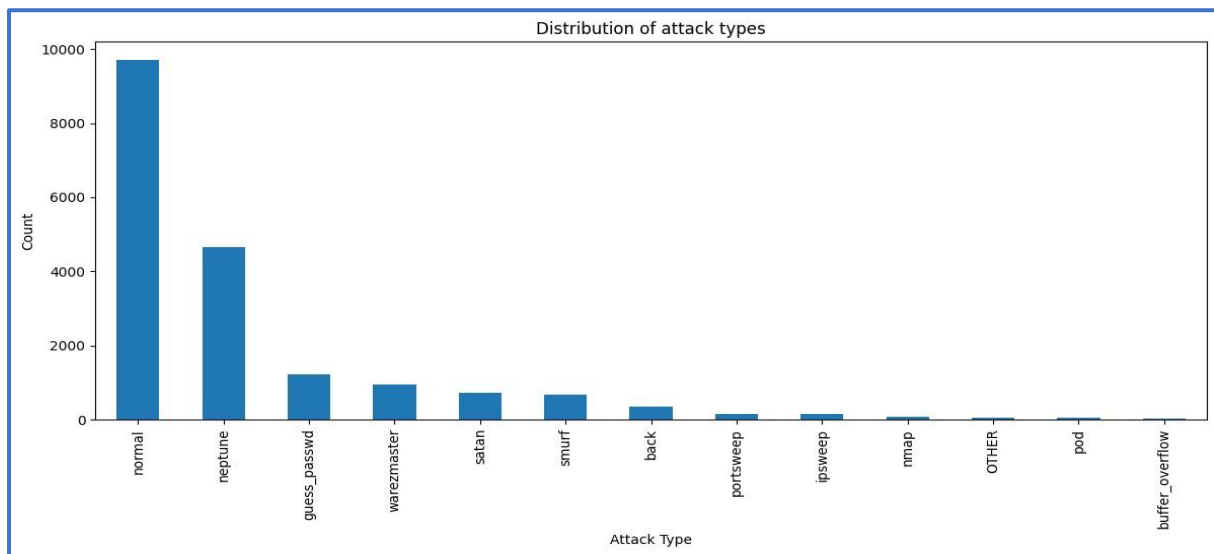


Figure 3: Count plots confirmed the imbalance between attack classes Testing Data.

Confusion Matrices

The confusion matrices highlighted differences in classifier performance. Logistic Regression misclassified several minority attacks, whereas Random Forest and Decision Tree provided stronger separation between normal and attack categories. This reinforced the effectiveness of tree-based models.

The confusion matrices provide insight into classification performance. As shown in *Figures 5 and 6*, Logistic Regression misclassified several minority attacks, while Random Forest and Decision Tree produced stronger separations between normal and attack categories.

Visualizations:

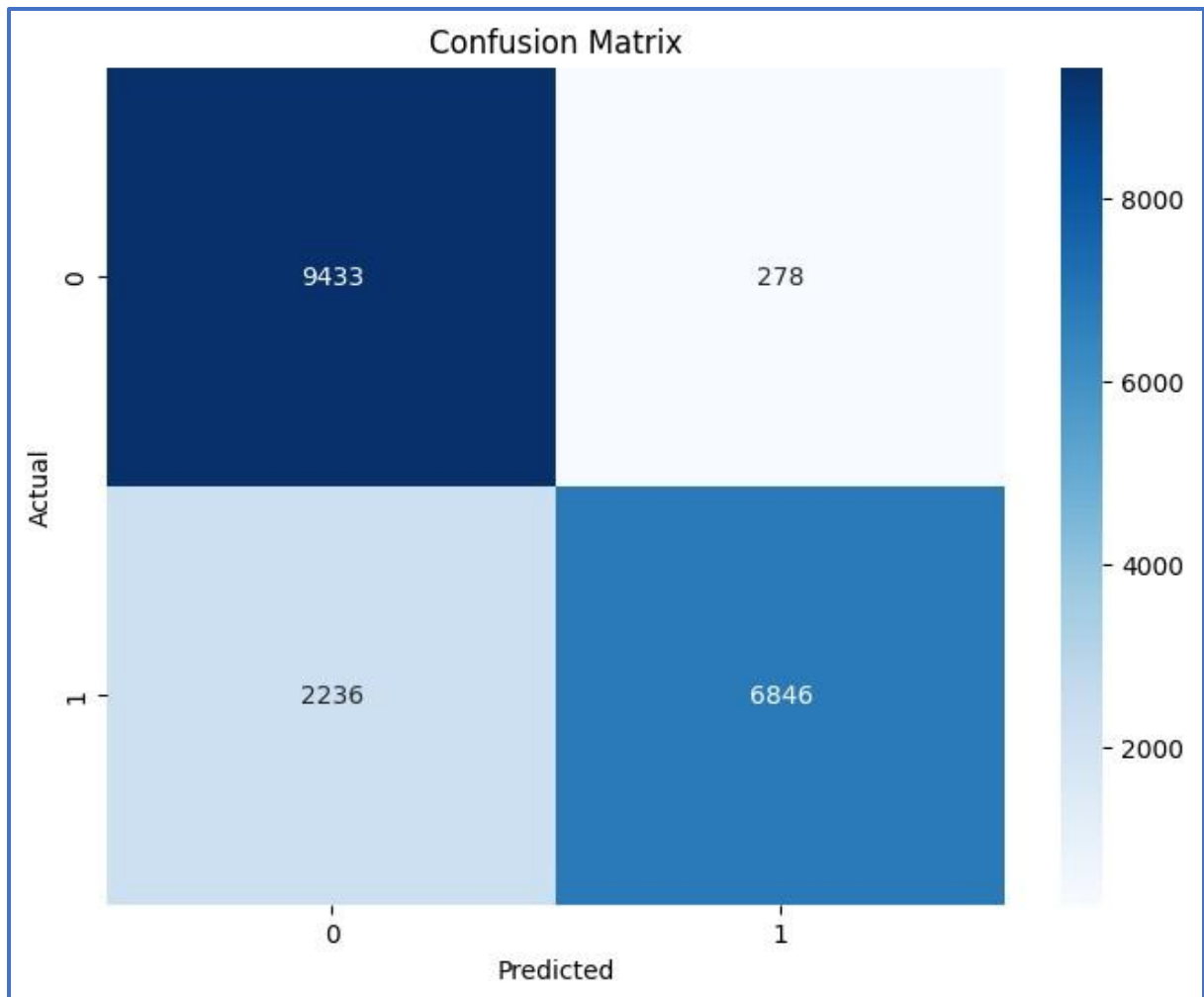


Figure 5: Confusion matrix for binary classification.

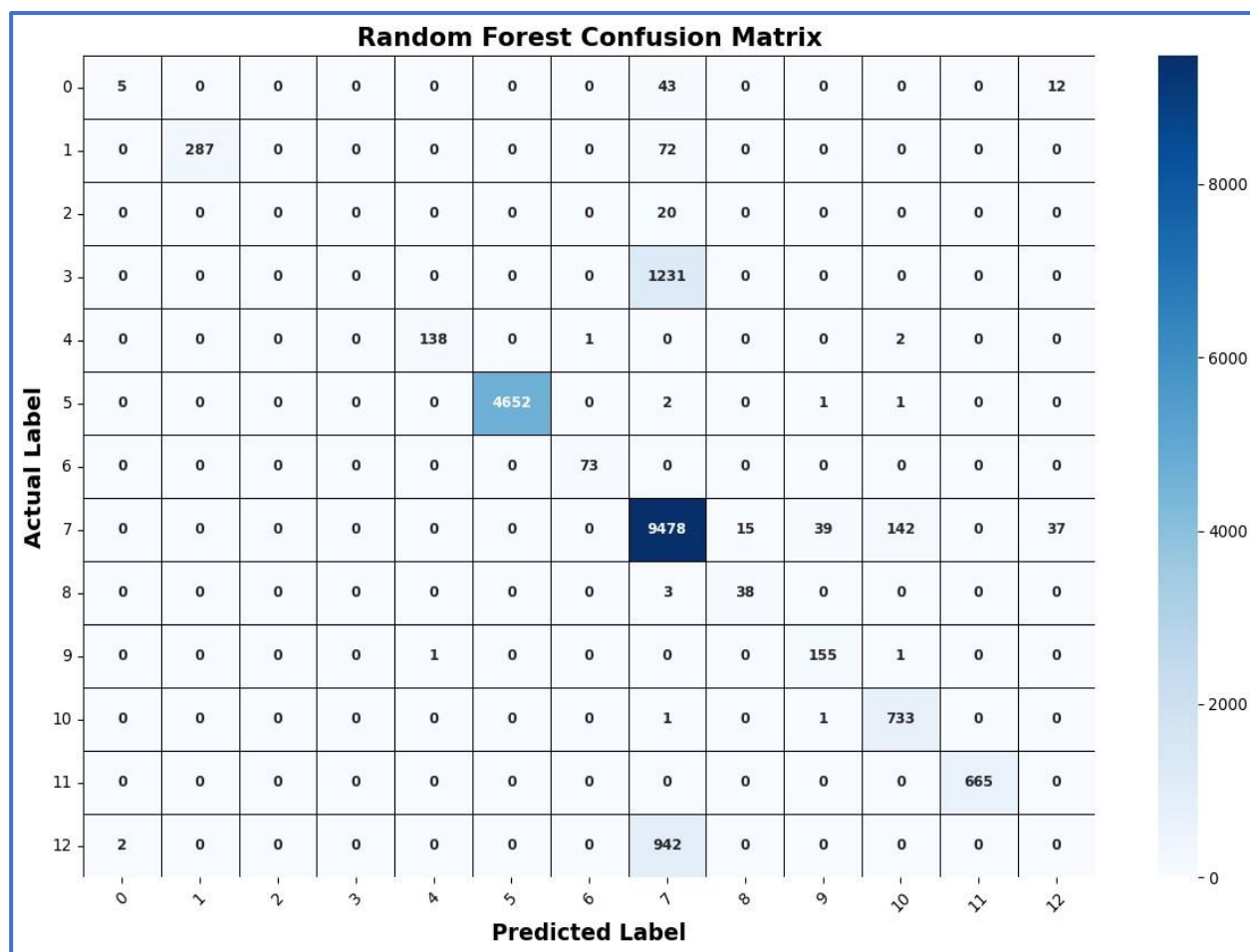


Figure 6: Confusion matrix for multiclass classification.

Results – Accuracy, Recall, Precision, F1-score

Logistic Regression achieved 97.11% accuracy on validation and 84.60% on test data. Random Forest reached 99.91% validation accuracy and 86.62% on test data. Decision Tree achieved 99.82% validation accuracy and 87.20% test accuracy. Macro F1-scores demonstrated that Random Forest and Decision Tree handled minority classes better.

Validation and test accuracies are presented in *Tables 1 and 2*. Logistic Regression achieved high validation accuracy (97%) but underperformed in detecting rare attacks. Random Forest and Decision Tree consistently achieved higher test accuracies (86–87%) and macro F1-scores, making them more reliable in real-world intrusion detection.

Figures 7 and 8 illustrate box plots comparing model accuracies for binary and multiclass classification, confirming the superior stability of tree-based models.

Model	Validation Accuracy	Test Accuracy	Macro F1-score For test accuracy
Logistic Regression	97.11%	84.60%	0.84
Random Forest	99.91%	86.62%	0.86
Decision Tree	99.82%	87.20%	0.87

Table 1: Model performance comparison table for Binary class classification

Model	Validation Accuracy	Test Accuracy	Macro F1-score For test accuracy
Logistic Regression	97.07%	84.22%	0.54
Random Forest	99.85%	86.33%	0.65
Decision Tree	99.84%	82.90%	0.59

Table 2: Model performance comparison table for Multiclass classification

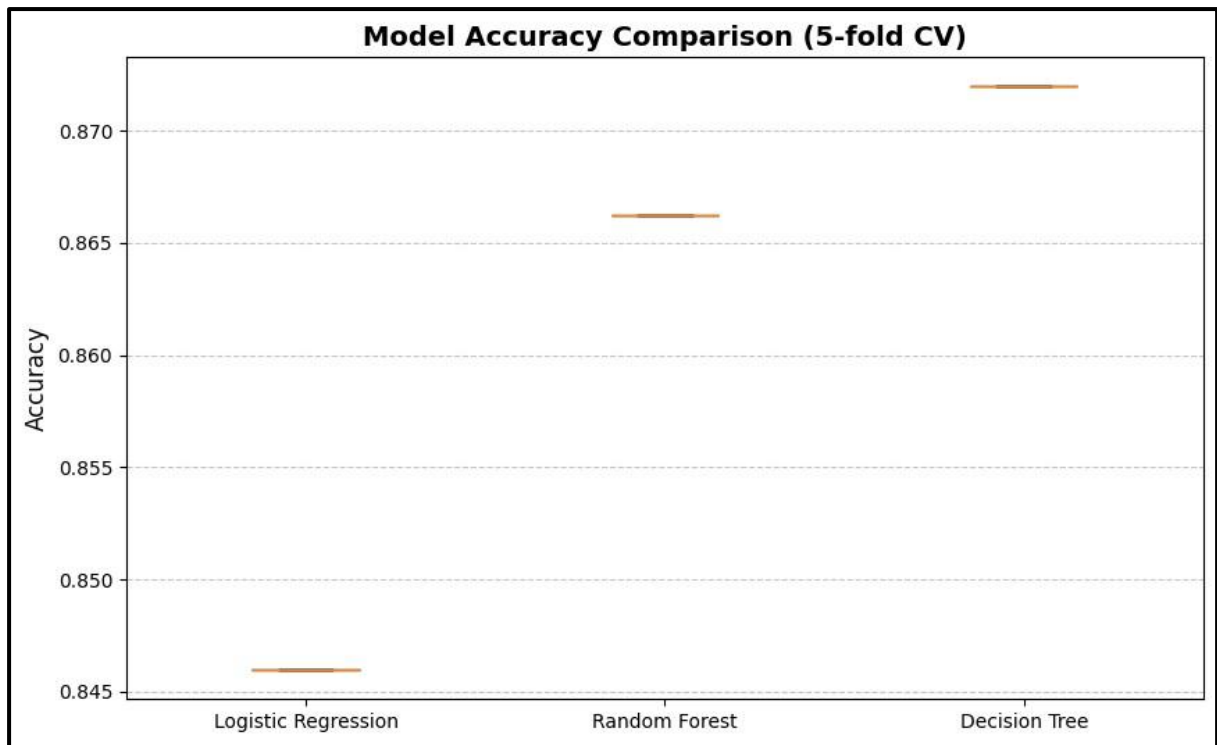


Figure 7: Box plot of model accuracies (binary classification)

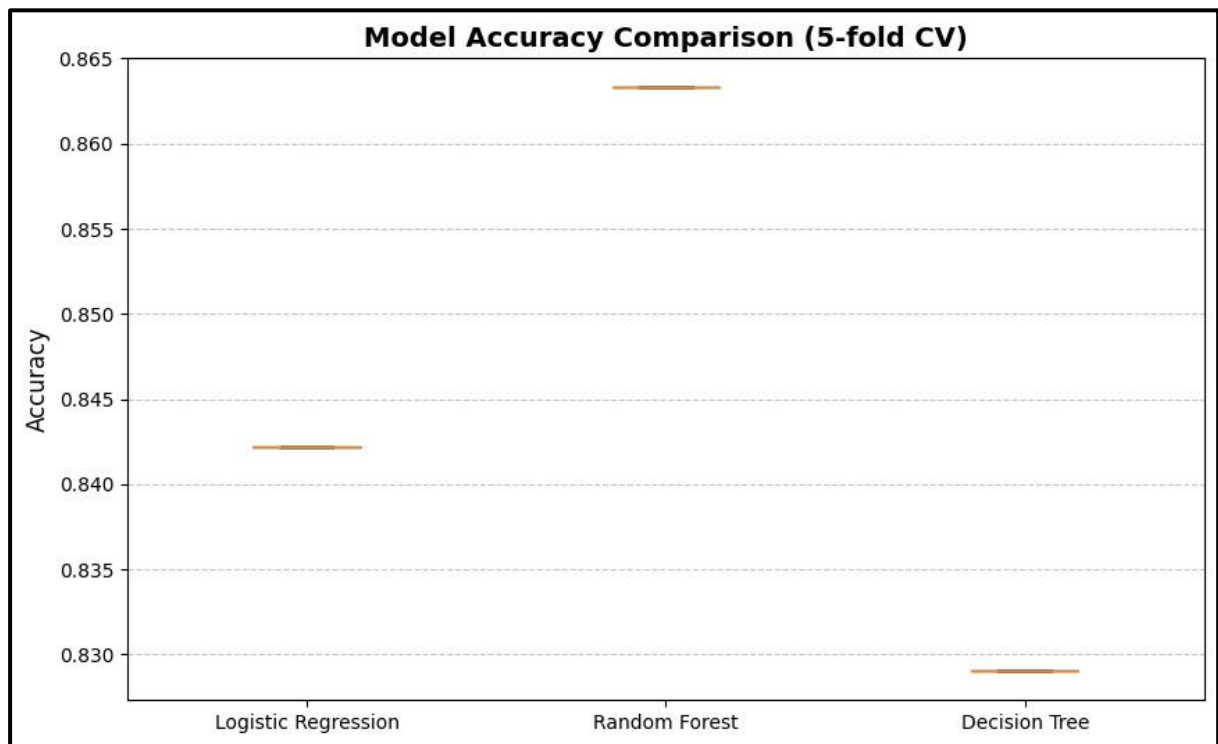


Figure 8: Box plot of model accuracies (multiclass classification)

ROC-AUC Curves

The ROC-AUC scores revealed that Random Forest and Decision Tree achieved values above 0.85, showing superior classification ability. Logistic Regression performed slightly lower, particularly in rare classes. ROC plots visually demonstrated the robustness of ensemble models in distinguishing normal from attack traffic.

The ROC-AUC scores further reinforce these findings. As shown in **Figures 9 and 10**, Random Forest and Decision Tree models achieved AUC scores above 0.85, demonstrating strong capability to distinguish between normal and attack traffic. Logistic Regression lagged slightly, particularly in handling rare classes.

Visualizations:

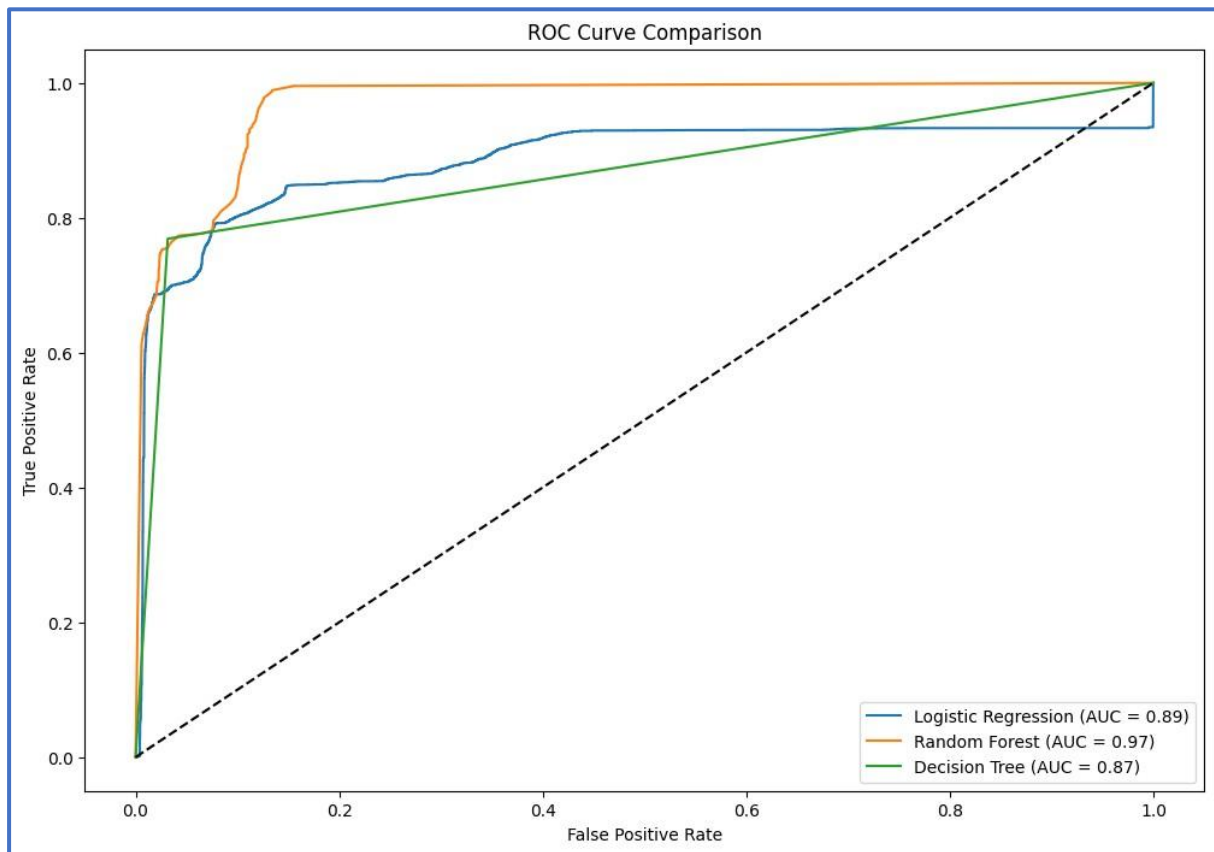


Figure 9: ROC-AUC curves for binary classification.

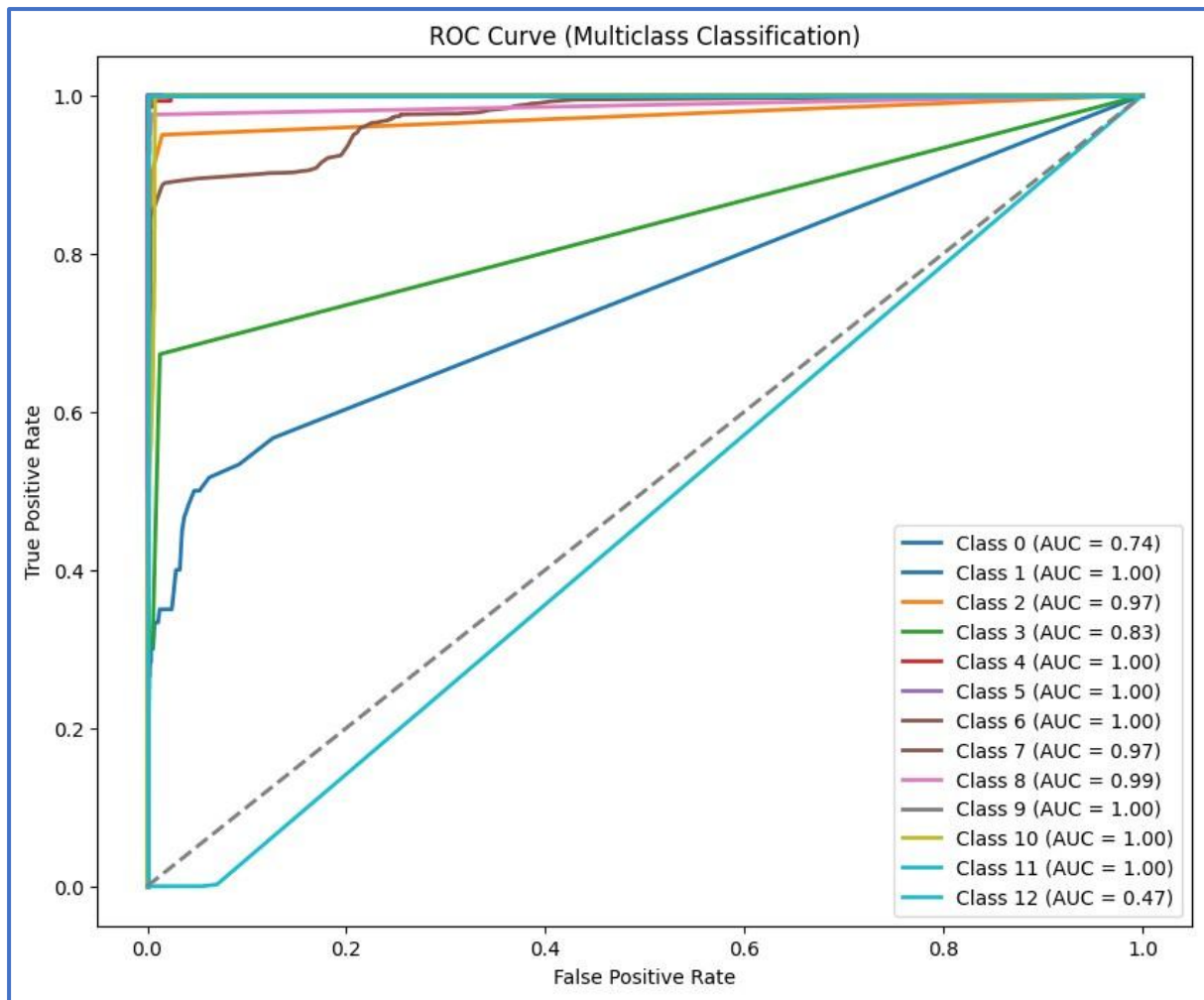


Figure 10: ROC-AUC curves for multiclass classification.

Discussion

The experiments revealed that Logistic Regression, while efficient, failed to detect rare classes reliably. Random Forest and Decision Tree showed better balance between accuracy and fairness across categories. The SMOTE-Tomek approach contributed significantly to improving minority class recognition.

Evaluation using macro F1-scores ensured that minority classes were not overshadowed by dominant categories. The findings suggest that ensemble models are better suited for IDS applications in highly imbalanced datasets.

The experiments conducted with the NSL-KDD dataset highlight several critical observations regarding the design and deployment of intrusion detection systems. First, the dominance of majority classes such as “Normal” and “Neptune” creates a bias in standard classifiers, leading to under-detection of minority but severe attack types like “Buffer Overflow” or “Guess Password.” This imbalance directly influences system robustness, as attackers often exploit rare vulnerabilities to evade detection.

The **SMOTE–Tomek resampling** technique played a crucial role in mitigating this imbalance. By generating synthetic samples of rare classes while cleaning ambiguous samples, the dataset became more representative of real-world attack scenarios. The improvement in macro F1-scores across Random Forest and Decision Tree confirms the effectiveness of this strategy.

Another key observation is the difference in classifier performance. Logistic Regression, despite being computationally efficient, failed to model nonlinear relationships between features and attack categories, resulting in poor recognition of minority classes. In contrast, tree-based models such as Random Forest and Decision Tree proved more adaptable by capturing nonlinear feature interactions. Notably, **Random Forest showed more stable performance across validation and testing**, while Decision Tree achieved slightly higher test accuracy.

The discussion also emphasizes the importance of **evaluation metrics**. Relying solely on accuracy can be misleading in imbalanced datasets, as high accuracy may still coincide with poor detection of minority classes. By using **macro F1-scores and ROC-AUC curves**, the evaluation captured fairness across categories and validated the models’ discriminatory power.

Conclusion

This project successfully designed and implemented an AI-powered intrusion detection system, Sentinel Net, leveraging the NSL-KDD dataset. Through a structured methodology involving preprocessing, hybrid resampling, exploratory analysis, and classifier comparison, the system demonstrated strong capabilities in identifying both common and rare attack types.

The results revealed that Random Forest and Decision Tree outperform Logistic Regression, particularly in handling data imbalance. Random Forest provided consistent macro F1-scores, while Decision Tree achieved slightly higher test accuracy. Both models proved suitable for real-world IDS tasks, where fairness across categories is as important as overall accuracy.

The work also highlights the significance of combining resampling techniques with ensemble models, which together enhance detection performance in highly imbalanced environments. By focusing on macro F1-scores, this study ensured that minority attacks were not overshadowed by dominant categories, thus aligning the system with practical cybersecurity requirements.

In conclusion, Sentinel Net demonstrates the potential of classical machine learning approaches combined with data balancing to deliver reliable, scalable, and interpretable intrusion detection solutions.

Future Works

While Sentinel Net achieved promising results, several directions exist for further improvement:

1. **Deep Learning Integration**

Advanced models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs/LSTMs) could be employed to capture spatial and temporal dependencies in network traffic. This may enable detection of more sophisticated attacks.

2. **Testing on Diverse Datasets**

Extending evaluation to larger and more recent datasets such as CIC-IDS-2017 or UNSW-NB15 would strengthen generalizability and confirm scalability across varied network environments.

3. **Real-Time Deployment**

Deploying Sentinel Net in a real-time monitoring environment could validate its practical usability. This would involve handling streaming network data, latency constraints, and integration with existing security infrastructure.

4. **Ensemble Deep Learning Methods**

Combining classical ensemble techniques with deep learning architectures could further improve robustness, balancing interpretability and predictive power.

5. **Feature Selection and Dimensionality Reduction**

Techniques such as PCA (Principal Component Analysis) or Autoencoders could be applied to reduce dimensionality, improving computational efficiency without compromising accuracy.

6. **Explainability and Interpretability**

Future systems should also focus on explainable AI (XAI) methods, which can help security analysts understand why a particular attack was detected or missed. This is essential for trust and adoption in enterprise environments.

Overall, these directions aim to transform Sentinel Net from a research prototype into production-ready intrusion detection system, adaptable to evolving cyber threat landscapes.